

* Loops

⇒ Loops in python are used to execute a block of code repeatedly until a certain condition is met.

Python mainly provides two types of loops:

- ① For loop
- ② While loop
- ③ Nested loops
 - For loop inside For loop
 - For loop inside while loop
 - While loop inside while loop
 - While loop inside For loop

1) For loop →

- For loop with range
- For loop with sequence

Syntax:

```
for i in sequence-name: # i Iterative variable  
    statements
```

Ex: name = "Kiran"

```
for i in name:
```

```
    print(i)
```

o/p k print (name)

i kitan

& we want to print position of the
a each character in the name w,
n use enumerate keyword.

```
for i, v in enumerate(name):
```

```
    print(i)
```

```
    print(f"pos = {i}, char = {i[i]}")
```

o/p (0, 'k')

(2, '&')

pos = 0, char = k

pos = 2, char = &

(1, 'i')

(3, 'a')

pos = 1, char = i

pos = 3, char = a

(4, 'n')

pos = 4, char = n

another

p = 0

```
for i, v in name:
```

```
    print(f"pos = {p}, value = {v}")
```

```
    p = p + 1
```

o/p

pos = 0, value = k

pos = 1, value = i

pos = 2, value = &

pos = 3, value = a

pos = 4, value = n

• for loop with range

initialization

condition

increment / decrement

range (start value, stop value, step size)

range (i=0, i < 11, i=i+1)

default start value : 0

step size : 1

Syntax :

for i in range():

statements

Ex: for i in range(1, 10, 1):

print (" i = ", i)

print ("Hello")

<u>O/P</u>	i = 1	i = 3	i = 5	i = 7	i = 9
	Hello	Hello	Hello	Hello	Hello
	i = 2	i = 4	i = 6	i = 8	i = 10
	Hello	Hello	Hello	Hello	Hello

Ex 1: Print numbers from 1 to 20

for i in range(1, 21, 1):
 print(i)

O/P

1

2

3

4

5

6

7

8

9

10

11

12

13

14

15

16

17

18

19

20

Ex 2: Print even numbers from 1 to 20

for i in range(0, 21, 2):
 print(i)

O/P

0 12

2 14

4 16

6 18

8 20

10

Ex: 3 Print odd numbers
from 1 to 100

for i in range(1, 100, 2):
 print(i)

O/P

1, 3, 5, 7, 9